

CODE SPORT



This month, we continue our discussion on testing machine learning applications, focusing on natural language processing (NLP).

In the last two columns, we looked at how to test machine learning (ML) applications. Now, we discuss the question, “How can we make our NLP model robust and production ready?” While NLP applications have made their mark in real-world deployments, there have also been many cases of NLP models having made glaring mistakes on trivial examples.

A well-known NLP task is Natural Language Inference (NLI). In this task, you are given a premise statement and a hypothesis statement. The task is to predict whether the hypothesis is entailed by the premise (if the premise is true, the hypothesis is also true), contradicted by the premise (if the premise is true, then the hypothesis is false) or neutral (where it is not possible to determine whether the hypothesis is true, given that the premise is true). Consider the following example.

Premise: There are two children building sand castles on the beach.

Hypothesis: Two children are playing on the beach.

In this case, the hypothesis is entailed by the premise. If you consider an alternative hypothesis: “Two adults are playing on the beach,” the relation between the premise and hypothesis is contradiction. If the hypothesis is, “A boy and a girl are playing on the beach,” then the relation between the premise and hypothesis is neutral.

Natural Language Inference is an important problem in NLP since it can help to verify facts, answer questions, etc. Consider this example.

Premise: John killed Mary.

Hypothesis: Mary killed John.

Even a five-year-old child with limited language comprehension would be able to determine that the relationship between the given premise and

hypothesis is contradiction. However, many of the state-of-art NLP systems end up mispredicting this pair as textual entailment. In fact, one state-of-art NLP system mispredicted this case with a confidence level of 92 per cent.

Well, you may be thinking that since Natural Language Inference is a complex task, it is possible that NLP systems made this mistake. Let us consider another NLP task, which is much simpler and mature, namely, sentiment analysis. Given a sentence, the task is to classify it as having either positive or negative sentiment. This is a simple binary classification task. There are a number of NLP vendors, including Google and Microsoft, that offer sentiment analysis as a service.

Let us try the following sentence on a sentiment analysis system, “A great white shark bit me on the beach.” Many of the sentiment analysis systems including that of Google Cloud-NLP, Microsoft’s Text Analytics services or even the Stanford Deep Learning based sentiment model predict this as carrying a high positive sentiment. However, even a child can determine that this sentence does not carry a positive sentiment.

Why do state-of-art NLP systems make such silly mistakes? Well, the answer lies in the fundamental design principles of modern NLP systems. The NLP tasks we discussed, such as sentiment analysis or natural language inference, are text classification problems. The classifier can typically either be a standard machine learning classifier or a neural network classifier using the supervised learning paradigm. Such classifiers are based on associative learning instead of causal learning. Consider a sentiment analysis system for movie reviews. A user sees a movie that he may like a lot, in which case, he writes a review